

ADHEESH TRIVEDI

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4TH YEAR, BS COMPUTER SCIENCE & ENGINEERING, IISER BHOPAL, INDIA

EDUCATION

IISER Bhopal | GPA: 9.34/10 (3.74/4)

2022 – 2026

B.S. | Computer Science & Engineering

Bhopal, India

Discrete Mathematics
Probability & Statistics
Complex Variables
Group Theory
Real Analysis
Linear Algebra
Multivariable Calculus
Applied Optimization
Theory of Computation

Signals & Systems
Machine Learning
Computer Vision
Artificial Intelligence
Database Systems
Introduction to C Programming
Data Structures & Algorithms
Algorithms (Advanced)
Software Modelling & Verification

Modern Cryptography
Information Theory & Coding
Computer Organization
Network Sciences
Operating Systems
Computer Networks
Advanced Programming
Compiler Design

RESEARCH PROJECTS



Practical study on embeddings between KS and LTS

Dec 2025 – Apr 2026

Mentor: [Dr. Arpit Sharma](#), PhD. [Shonak Saha](#)

- Built a Python framework unifying model checking across multiple backends (NuSMV, Spin, Storm) by embedding LABELLED TRANSITION SYSTEMS into KRIPKE STRUCTURES.
- Benchmarked and studied the feasibility of explicit model checking in embedded models.



On Robust Coloring of Graphs and ETH

July 2025 – Jan 2026

Mentor: [Dr. Prafullkumar Tale](#)

- Studied a relaxation of proper coloring in which edges with same color are associated with some cost.
- Studied the EXPONENTIAL TIME HYPOTHESIS, and ETH-Preserving reductions.
- Investigated treewidth; applied tree decompositions to dynamic programming on hard problems.



Scientific Tool for Bridging Model Checking Ecosystems

Dec 2024 – Dec 2025

Mentor: [Dr. Arpit Sharma](#), PhD. [Shonak Shaha](#)

- The Model Checking ecosystem remains fragmented, with various tools and frameworks lacking seamless interoperability specifically between action based and state based model checking.
- Developing high-performance converters between action-labeled model checkers ([CADP](#)) and ([mCRL2](#)) and state-labeled model checkers ([PRISM](#)) and ([Storm](#)) to enable cross-ecosystem model checking.
- Emphasis on parser design, efficient graph transformations, and AST driven translation, written in C++.



Reading Project on Graph Theory

Jan 2024 – Apr 2024

Mentor: [Dr. Prafullkumar Tale](#)

- Studied *A First Look at Graph Theory* (Clark & Holton) with problem solving.
- Reinforced concepts: Graphs & their types, connectivity, traversals, matching problem, planarity.

SKILLS

Python • C++ (CMake, GoogleTest) • Git & GitHub • $\text{\LaTeX}$ • typst • Linux Shell • CI/CD • React (basic) • Rust (basic)

TECHNICAL PROJECTS

-  **Nifty Network Analysis & Visualization Dashboard** Oct 2025 – Nov 2025
– Analyzed relationships among major Nifty companies and summarized insights in an interactive dashboard and formal report.
– Delivered clear visual findings on market structure, influential firms, and cluster patterns for course project.
-  **Extensively Customizable Exam Scheduler using Graph Coloring** Oct 2024 – Jan 2025
Vivek Kumar, Rahul Jana, Ayushman Shaha, [Dr. Prafullkumar Tale](#)
– The exam schedules for 2,000+ students in my college were being created manually, a lot of effort.
– Developed an engine that assigns exams to time slots & halls minimizing same-day conflicts ($\leq 24h$).
– Uses graph coloring and randomized optimization heuristics for reaching close to optimal scheduling.
– Supports multi-hall allocation when enrollment exceeds single capacity.
-  **Texture Classification & Face Clustering for Image Search** Sep 2024 – Nov 2024
– Addresses the common challenge of navigating through a large collection of images,
– Enables the user to efficiently filter and search for images by face or texture.
– Implemented batching to control GPU memory and multiple texture descriptors for retrieval precision.
-  **GUI Developer Intern** Feb 2024 – Aug 2024
Mentor: [Prof. Vardharajan Srinivasan](#) AITG Labs
 – Led integration of DFT engines ([Octopus](#), [GPAW](#), [NWChem](#)) within a Python toolkit for photo-induced phenomena simulations.
– Automated submission workflows and job orchestration on HPC clusters.
-  **Context aware Bag-of-Words Chatbot** Oct 2021 – Dec 2021
– Built a context-aware chatbot supporting tasks like to-do lists, word definitions, note taking, make jokes and time queries across timezones.
– Trained a sequential neural network in TensorFlow on a custom dataset tailored to project requirements.
-  **General Purpose Discord Bot** Mar 2021
– Python project that utilized PostgreSQL to implement feature-rich Discord bot.

TALKS & TEACHING

- Teaching Assistant - Theory of Computation** Aug 2025 – Dec 2025
Instructed by [Dr. Arpit Sharma](#)
-  **Fast Matrix Multiplication Algorithms** 1 Sept 2023
Math Club @ IISER Bhopal

ACADEMIC ACHIEVEMENTS

- Secured **rank 29** (99.96 %ile) in GATE Data Science & Artificial Intelligence 2026
- Secured **rank 147** (99.93 %ile) in GATE Computer Science 2026
- Scored **92%** in 10th board (ICSE) and **91.2%** in 12th board (CBSE)

LEADERSHIP & ENGAGEMENT

- Club Coordinator** May 2024 – May 2025
Coding club @ IISER Bhopal
- Hackathon Organizer -- Armacode 0** Jan 2024 – Apr 2024
IISER Bhopal x IIIT Bhopal

CO-CURRICULAR

Music (Guitar) • Competitive Programming • Speed Typing